INCH-POUND
A-A-59440/4D
02 July 2020
SUPERSEDING
A-A-59440/4C
06 December 2018

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

COCK-VALVE, DRAIN, GROUND KEY,150 PSI, BRASS, STEAM PRESSURE, NPTF-1 THREADS

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring cock-valve, drain, ground key, 150 psi, brass, steam pressure, NPTF-1 threads described herein shall consist of this document and the issue in effect of A-A-59440.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the PIN as shown in the following example (see NOTES).

AA59440	/04	-	014
CID number	Applicable CID spec sheet (two digits)		Dash number (see figure 1)

SALIENT CHARACTERISTICS.

Interface and physical dimensions. Cock-valves, drain, ground key, 150 psi, brass, steam pressure, NPTF-1 threads supplied to this CID specification sheet shall be as specified herein (see figure 1).

Sizes. Nominal sizes of 1/8 inch through 1/2 inch with NPTF-1 threads.

Intended applications. This item is for use in steam systems where pressures do not exceed 150 psig (10.3 bar) at 351°F (177.2°C) and the liquid or gas is compatible with brass. Not for use in potable water systems.

Permissible variations in dimensions. Figure 1 lists the permissible variation in dimensions for each size. The external dimensions of the cock-valve should not limit procurement unless the end item applications restrict these dimensions.

Optional designs. Figure 1 is for identification purposes only, it is not intended to restrict designs or shapes not dimensioned.

Valve bodies. Valve bodies shall be provided with hexagonal wrenching surfaces.

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Threads. Threads are dryseal NPTF-1 in accordance with ASME B1.20.3 per A-A-59440.

Preferred valve body material. Preferred valve body material is brass in accordance with ASTM B61.

Material.

Material body, handle, stem, and screw. Body, handle, stem, and screw shall be brass in accordance with the following commercial standards:

ASTM B16/B16M	ASTM B121/B121M	ASTM B369	SAE AMS4845
ASTM B22	ASTM B124/B124M	ASTM B505/B505M	SAE AMS4855
ASTM B30	ASTM B148	ASTM B584	SAE AMS4860
ASTM B36/B36M	ASTM B176	ASTM B763	SAE AMS4862
ASTM B61	ASTM B271/B271M	ASTM B770	SAE AMS4890
ASTM B62			

NOTE:
1. Brass with lead contents shall not be used in potable water systems.
2. Class I and II ozone depleting substances (ODS) shall not be used in A-A-59440/4 or any referenced procedures.

Protective coating.

Brass: None.

Test pressures.

a. Shell: 300 psig (20.7 bar).
b. Seat: 150 psig (10.3 bar).

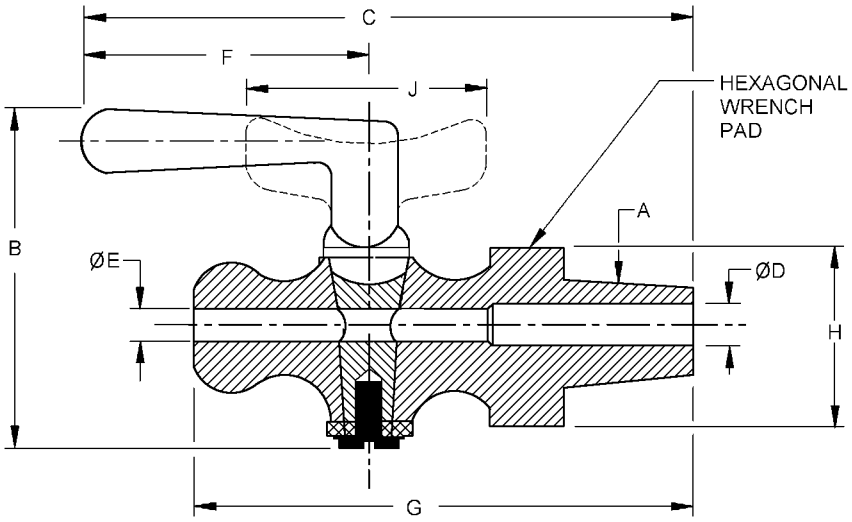


FIGURE 1. Cock, drain, ground key.

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Lever handle.

Dash number	A pipe thread NPTF-1	B max (mm)	C max (mm)	D diameter min (mm)	E diameter min (mm)	F max (mm)	G max (mm)	H across flats ±.015 (0.38 mm)
014	1/8-27	1.47 (43.7)	2.25 (57.2)	.16 (4.1)	.16 (4.1)	1.22 (30.1)	1.72 (2.5)	.50 or .56 (12.7 or 14.2)
015	1/4-18	1.66 (42.2)	2.56 (65.0)	.19 (4.8)	.16 (4.1)	1.41 (35.8)	1.97 (2.5)	.56 or .62 (14.2 or 15.7)
016	3/8-18	1.88 (47.8)	3.12 (79.3)	.25 (2.5)	.22 (5.6)	1.72 (43.7)	2.34 (59.4)	.69 or .81 (17.5 or 20.6)
017	1/2-14	2.44 (62.0)	3.94 (100.1)	.38 (9.7)	.28 (7.1)	2.16 (54.9)	2.88 (73.2)	.94 or 1.00 (23.9 or 25.4)

Tee handle.

Dash Number	A pipe thread NPTF-1	B max (mm)	D diameter min (mm)	E diameter min (mm)	G max (mm)	H across flats ±.015 (0.38 mm)	J max (mm)
018	1/8-27	1.47 (43.7)	.16 (4.1)	.16 (4.1)	1.72 (2.5)	.50 or .56 (12.7 or 14.2)	.97 (24.6)
019	1/4-18	1.66 (42.2)	.19 (4.8)	.16 (4.1)	1.97 (2.5)	.56 or .62 (14.2 or 15.7)	1.09 (27.7)
020	3/8-18	1.88 (47.8)	.25 (2.5)	.22 (5.1)	2.34 (59.4)	.69 or .81 (17.5 or 20.6)	1.22 (31.0)

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified tolerances are ±.010 inch (0.25 mm).

FIGURE 1. Cock, drain, ground key - Continued.

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

COMMERCIAL ITEM DESCRIPTION

A-A-59440 - Cock-Valves and Valves, General Requirements for

(Copies of these documents are available online at https://quicksearch.dla.mil/ or https://assist.dla.mil/).

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Other Publications

ASME INTERNATIONAL

ASME B1.20.3 - Dryseal Pipe Threads (Inch)

(Copies of these documents are available online at https://www.asme.org)

ASTM INTERNATIONAL

ASTM B16/B16M	-	Rod, Brass, Free-Cutting, Bar and Shapes for use in Screw Machines
ASTM B22	-	Castings, Bronze, for Bridges and Turntables
ASTM B30	-	Copper Alloys in Ingot Form
ASTM B36/B36M	-	Plate, Brass, Sheet, Strip, and Rolled Bar
ASTM B61	-	Castings, Steam or Valve Bronze
ASTM B62	-	Metal Castings, Composition Bronze or Ounce
ASTM B121/B121M	-	Plate, Leaded Brass, Sheet, Strip, and Rolled Bar
ASTM B124/B124M	-	Copper and Copper Alloy Forging Rod, Bar, and Shapes
ASTM B148	-	Aluminum-Bronze Sand Castings
ASTM B176	-	Die Castings, Copper-Alloy
ASTM B271/B271M	-	Copper-Base Alloy Centrifugal Castings
ASTM B369	-	Copper-Nickel Alloy Castings
ASTM B505/B505M	-	Castings, Copper Alloy, Continuous
ASTM B584	-	Castings, Copper Alloy Sand, for General Applications
ASTM B763	-	Copper Alloy Sand Castings for Valve Application
ASTM B770	-	Copper-Beryllium Alloy Sand Castings for General Applications

(Copies of these documents are available online at https://www.astm.org/ or from the ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

SAE INTERNATIONAL

SAE AMS4845	-	Tin Bronze Castings, Sand and Centrifugal 87.5CU - 10SN - 2ZN as Cast
SAE AMS4855	-	Leaded Red Brass, Sand and Centrifugal Castings 85CU - 5.0SN - 5.0PB - 5.0ZN as Cast
SAE AMS4860	-	Manganese Bronze, Sand and Centrifugal Castings 58CU - 39ZN - 1.2FE - 1.0AL - 0.80MN as Cast
SAE AMS4862	-	Manganese Bronze, Sand and Centrifugal Castings 63CU - 24ZN - 6.2AL - 3.8MN - 3.0FE High Strength, as Cast
SAE AMS4890	-	Copper-Beryllium Alloy Castings 97CU - 2.1BE - 0.52CO - 0.28SI Solution Heat Treated (TBOO)

(Copies of these documents are available on line at https://www.sae.org/from SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001, and Tel: 877-606-7323 [inside USA and Canada] or 724-776-4970 [outside USA], email at CustomerService@sae.org.)

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Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

MFR's CAGE	MFR's name and address
01276	Eaton Aeroquip Inc. 14615 Lone Oak Rd. Eden Prairie, MN 55344 952-294-2032/888-258-0222 jeffenoe@eaton.com
93061	Parker-Hannifin Corp Brass Products Division 300 Parker Drive Otsego, MI 49708-1472 260-636-2104 dwsanders@Parker.com
20969	Essex Brass Corp 23500 Pinewood Warren, MI 48091-3122 586-757-8200 bpushman@essexbrass.com

Military standard (MS) supersession data. This CID replaces the following MS as shown in table I.

TABLE I. MS supersession data.

CID PIN	Superseded MS
AA59440/4-014	MS35785-1
AA59440/4-015	MS35785-2
AA59440/4-016	MS35785-3
AA59440/4-017	MS35785-4
AA59440/4-018	MS35785-5
AA59440/4-019	MS35785-6
AA59440/4-020	MS35785-7

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Part number P/N supersession data. These CID specification sheet PINs supersede the following MFR's P/N's as shown in table II. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. P/N supersession data.

Dash number (see figure 1) AA59440/4	MFR's CAGE	MFR's P/N 1/
014	20969	SC63L
015	20969	SC64L
016	20969	SC14
017	20969	SC15
018	20969	SC63T
019	20969	SC64T
020	20969	SC8

1/ The manufacturer's P/N shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see marking paragraph of A-A-59440.

National stock number (NSN). The following is a list of NSN's assigned which correspond to this CID specification sheet. The list in table III is for information only and may not be indicative of all possible NSN's associated with the CID specification sheet. For up to date information on assigned NSN's, please contact the aforementioned DLA Land and Maritime office listed in A-A-59440.

TABLE III. NSN's.

Dash number (see figure 1) AA59440/4	NSN
015	4820-00-197-4984
019	4820-00-057-0694

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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MILITARY INTERESTS:

Custodians:
Army - MI
Navy - SH
Air Force - 71
DLA - CC

Review activities:
Army - AT, CR4
Navy– AS, SA

CIVIL AGENCY AND COORDINATING ACTIVITIES:

GSA - FSS

Preparing activity:

DLA - CC

(Project 4820-2020-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organization and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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